AI FOR SCIENCE:
PUTTING PHYSICS INTO LEARNING

Introductory lecture

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What AI has made possible

Problems solved in the last few years

Protein structure

3D structure from the sequence alone, close to experimental accuracy

Molecular simulation

quantum-chemistry accuracy at force-field cost

Weather forecasting

10-day global forecast in a minute, as good as the best physics models

Exoplanet detection

a network finds a planet hidden in 4 years of Kepler light curves

Materials discovery

hundreds of thousands of new crystals predicted stable

Maths & algorithms

faster matrix multiplication, new combinatorial constructions

Problems solved in the last few years

Protein structure

3D structure from the sequence alone, close to experimental accuracy

AlphaFold 2 (2020)

Molecular simulation

quantum-chemistry accuracy at force-field cost

Deep Potential (2018)
MACE (2022)

Weather forecasting

10-day global forecast in a minute, as good as the best physics models

GraphCast (2023)

Exoplanet detection

a network finds a planet hidden in 4 years of Kepler light curves

Shallue & Vanderburg (2018)

Materials discovery

hundreds of thousands of new crystals predicted stable

GNoME (2023)

Maths & algorithms

faster matrix multiplication, new combinatorial constructions

AlphaTensor (2022)
FunSearch (2023)

AlphaFold: Nature 596, 583 (2021) · DeePMD: PRL 120, 143001 (2018) · MACE: NeurIPS (2022) · GraphCast: Science 382, 1416 (2023) · Kepler-90i: AJ 155, 94 (2018) · GNoME: Nature 624, 80 (2023) · AlphaTensor: Nature 610, 47 (2022) · FunSearch: Nature 625, 468 (2024)

What did AI improve?

Speed

an expensive calculation becomes cheap

Molecular simulation
DFT → ML potential: 10^3–$10^6\times$ faster

Weather
hours on a supercomputer → one minute on one chip

Search

explore a huge, high-dimensional space

Materials
combinatorial space of compositions and structures

Maths & algorithms
space of programs, too large to enumerate

Unknown map

learn a relation no theory gives in a usable form

Protein structure
sequence → 3D structure: seconds to hours instead of months to years of experiments

Finding signals

rare, weak patterns in huge datasets

Exoplanets
trained on $\sim 15\,000$ labelled Kepler signals, then searched new stars: found Kepler-90i, the 8th planet of its system

Why did these work?

Data

large curated databases built over decades (Protein Data Bank, weather reanalyses), or a simulator that produces labels on demand (DFT)

A way to check

we can tell whether a prediction is right: blind tests (CASP), comparison with what happened, a reference calculation, a proof

Known structure

symmetries, locality, conservation laws built into the model (rotation symmetry in AlphaFold and MACE)

- All of these models use prior knowledge of the problem, in addition to data
- In science we usually know a lot beforehand: **equations, symmetries, conservation laws**
- **This lecture: how to put this knowledge into a learned model**

Science for AI: the other direction

Concept from science		AI method
Langevin dynamics, non-equilibrium thermodynamics	→	**diffusion models** (2015, 2020)
Boltzmann distribution, spin models	→	energy-based models, Boltzmann machines
Monte Carlo (1953); Hybrid Monte Carlo from lattice QCD (1987)	→	MCMC and HMC for Bayesian inference
variational free energy, mean-field theory	→	variational inference, VAEs (ELBO $= -$free energy)
optimal transport (Monge, Kantorovich)	→	Wasserstein GANs, flow matching
group theory, symmetries	→	equivariant networks (Part II)

Many ML methods come from physics and statistics.

Sohl-Dickstein et al., ICML (2015) · Ho et al., NeurIPS (2020) · Metropolis et al., J. Chem. Phys. 21, 1087 (1953) · Duane et al., Phys. Lett. B 195, 216 (1987)

ML in 5 minutes

Learning = fitting a function

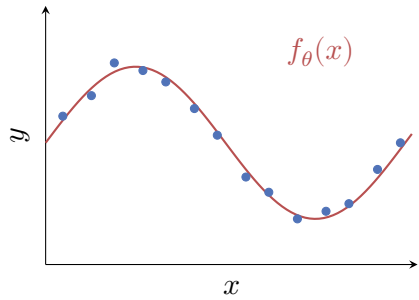
- Data: pairs (x_i, y_i), $i = 1 \dots N$
- Model: a flexible function $f_\theta(x)$ with parameters θ
- Loss: how wrong we are

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \ell(f_\theta(x_i),\, y_i)$$

- Optimisation: gradient descent

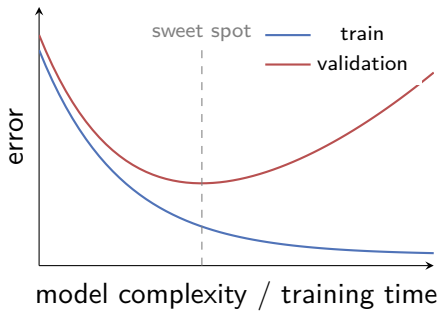
$$\theta \leftarrow \theta - \eta \, \nabla_\theta \mathcal{L}$$

- f_θ is often a **neural network** (universal approximation theorem)

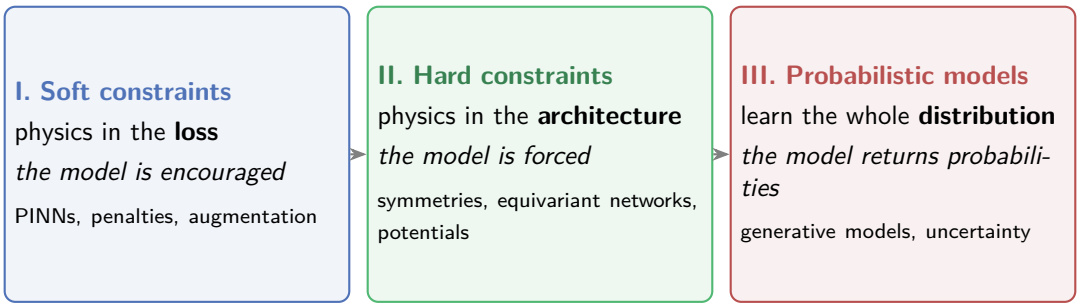


Generalisation and inductive bias

- Goal: perform well on **new** data, not on the training set
- Split data: **train** / **validation** / **test**
- Overfitting: the model memorises noise
- **Inductive bias**: the assumptions built into the model decide how it generalises, especially **outside the data**
- Physical knowledge is a strong inductive bias



Three ways to put physics into a model



Prior knowledge reduces the amount of data needed.

I. Soft constraints: physics in the loss

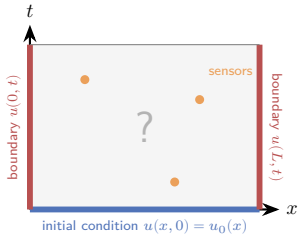
The idea: add physics to the loss

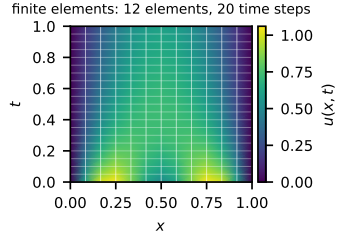
$$\mathcal{L}(\theta) = \underbrace{\frac{1}{N}\sum_i \left(f_\theta(x_i) - y_i\right)^2}_{\text{fit the data}} + \sum_k \lambda_k \underbrace{\mathcal{R}_k[f_\theta]}_{\text{violate the physics}}$$

- $\mathcal{R}_k \geq 0$, zero when the physics is satisfied
- Can be evaluated **anywhere**, not only where we have data
- The network architecture does not change: any model, any optimiser
- Physics acts as a **regulariser**: it pulls the model towards physical solutions
- But the constraint is only satisfied **approximately**, as a trade-off set by λ_k

The problem: solving a PDE with a neural network

$$\partial_t u = D\,\partial_x^2 u \quad x \in [0, L],\ t \in [0, T] \qquad u(x, 0) = u_0(x) \qquad u(0, t) = u(L, t) = 0$$

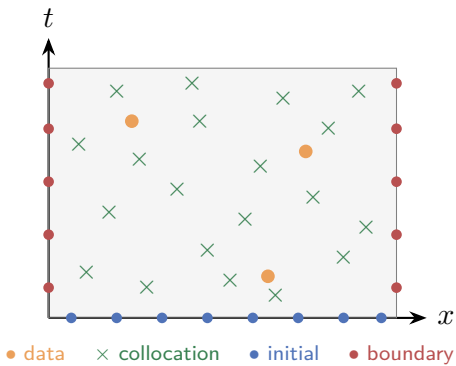




- **Known**: the equation, initial and boundary conditions, a few sensors
- **Unknown**: $u(x, t)$ everywhere inside

- Classical route (right): mesh + finite elements, solve on the grid
- Neural-network route: $u_\theta(x, t)$, mesh-free and differentiable
- But: no labels inside the domain. How do we train it?

The PINN solution: the equation becomes the training signal

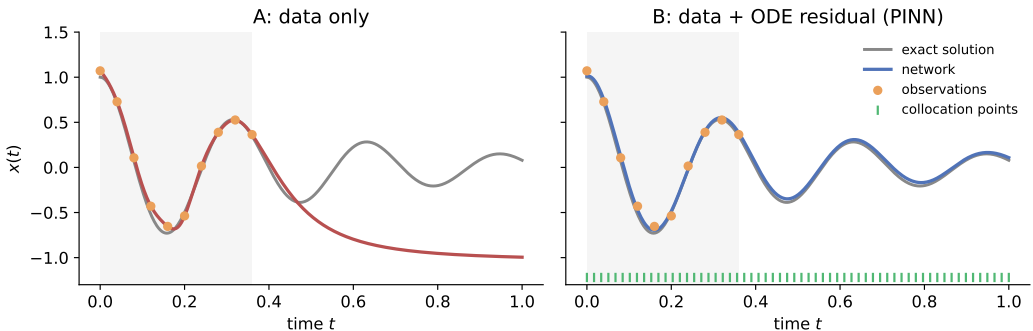


$$\mathcal{L}(\theta) = \sum_i \left|u_\theta(x_i, t_i) - u_i\right|^2 + \frac{1}{N_r}\sum_r \left|\partial_t u_\theta - D\,\partial_x^2 u_\theta\right|^2_{(x_r,t_r)} + \sum_j \left|u_\theta(x_j, 0) - u_0(x_j)\right|^2 + \sum_k \left|u_\theta(x_k^\partial, t_k) - g(t_k)\right|^2$$

- Derivatives of u_θ by **automatic differentiation**: exact, no mesh
- Collocation points are **free**: no labels needed, put them anywhere
- No data $\Rightarrow$ forward solver; with data $\Rightarrow$ **inverse problem** (e.g. learn D too)

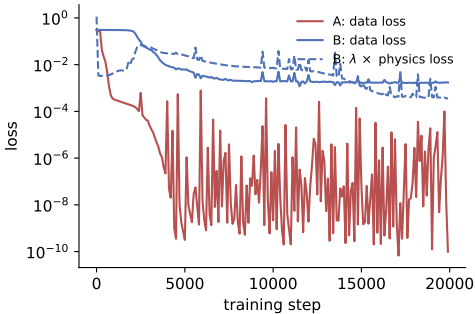
Lagaris, Likas & Fotiadis, IEEE TNN 9, 987 (1998) · Raissi, Perdikaris & Karniadakis, J. Comput. Phys. 378, 686 (2019)

Demo I: a damped oscillator, observed only at the beginning

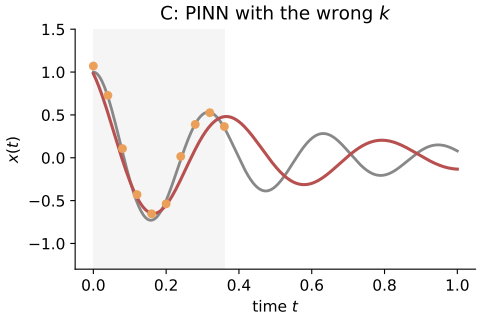


$\ddot{x} + \mu\dot{x} + kx = 0$, 10 noisy points in the grey region. **Same network**, only the loss differs: A: data only $\Rightarrow$ fits the points, wrong afterwards. B: + ODE residual on 60 collocation points $\Rightarrow$ follows the physics everywhere.

Demo I: two lessons



The data-only model has the **lowest training loss** (it memorises the noise) and the **worst** prediction.



A PINN with the **wrong** stiffness k fits the data and gives a smooth, plausible curve that is wrong afterwards. The result relies on the physics being correct.

Soft constraints beyond PDEs

Conservation and sum rules

- total charge $\sum_i q_i = Q$ in ML charge models
- $\int \rho(\mathbf{r})\, d\mathbf{r} = N_{\text{electrons}}$, f-sum rule
- probabilities sum to one

Energy–force consistency

- separate heads for E and $\mathbf{F}$, penalty $\|\mathbf{F} + \nabla E\|^2$

Known limits

- ideal-gas limit, short-range repulsion, dissociation limit

Kinetics

- detailed balance $\pi_i k_{ij} = \pi_j k_{ji}$ in learned rate matrices; non-negative rates

Simulation-based constraints

- the model output goes through a simulation before the loss: $\mathcal{L} = \|\mathrm{Sim}(f_\theta) - y_{\text{meas}}\|^2$
- gradients go through the simulator: **differentiable simulation**
- e.g. adjust a force model until simulated and observed trajectories match

Shape constraints

- monotonicity, convexity, bounds

Any equation, inequality or limit you trust can become a loss term.

Two more soft tools

Data augmentation

- Known symmetry? Show the model transformed copies of the data (rotated, permuted, translated)
- The model **learns** the symmetry, approximately and only near the data
- The soft version of what Part II does exactly

Neural operators

- Learn a map between **functions**: initial condition, parameters $\mapsto$ solution field
- Fourier neural operators, DeepONet (Li et al. 2021; Lu et al. 2021)
- Physics-informed versions add the PDE residual to the loss: one model for a whole family of problems

Li et al., ICLR (2021) · Lu et al., Nat. Mach. Intell. 3, 218 (2021)

Soft constraints: limitations, and when to use them

Limitations

- Choosing the weights λ_k: competing terms
- Stiff, ill-conditioned loss landscapes; PINNs can fail to train (Krishnapriyan et al. 2021; Wang et al. 2021)
- Spectral bias: slow to learn sharp or high-frequency features
- Constraint satisfied only approximately, only where evaluated
- Wrong physics $\Rightarrow$ wrong model, without warning

When to use them

- Known equations, scarce or noisy data
- Inverse problems: infer parameters or a missing term
- Approximate physics you don't want to impose exactly
- Quick to try: only the loss changes

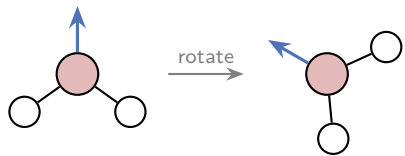
This week: *"What accuracy is lost by removing the physics?"* is one of the questions of the surrogate challenge.

Krishnapriyan et al., NeurIPS (2021) · Wang, Teng & Perdikaris, SIAM J. Sci. Comput. 43, A3055 (2021)

II. Hard constraints: physics in the architecture

Symmetries: invariance and equivariance

- Physical laws do not depend on how you label or orient things
  - translations, rotations, permutations of identical particles, time reversal, gauge
- **Invariance**: $f(g \cdot x) = f(x)$
  - energy $E(R\mathbf{r}) = E(\mathbf{r})$
- **Equivariance**: $f(g \cdot x) = g \cdot f(x)$
  - forces $\mathbf{F}(R\mathbf{r}) = R\,\mathbf{F}(\mathbf{r})$
- How do we get a model that satisfies this **exactly**?



E unchanged (invariant)
force rotates (equivariant)

Four routes to an exact symmetry

1. Invariant inputs

$x \;\rightarrow$ distances, angles $\rightarrow$ NN

describe the system only by invariants (symmetry functions, SOAP, ACE)

2. Canonicalise

$x \rightarrow$ standard pose $\rightarrow$ NN

align to inertia frame, sort atoms; or learn the frame

3. Average

$\bar{f}(x) = \frac{1}{|G|}\sum_g g^{-1} f(gx)$

symmetrise any model over the group or a set of frames

4. Equivariant layers

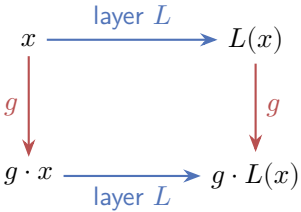
$x \rightarrow$ layer $\rightarrow \cdots \rightarrow$ layer

every layer commutes with the group (CNN, GNN, e3nn)

+ simple, cheap	any network, exact	any network, exact	most expressive, data-efficient
– may lose information (chirality), scalar outputs	discontinuous at degeneracies	cost $\propto$ number of group elements	complex, costly

Behler & Parrinello, PRL 98, 146401 (2007) · Kaba et al., ICML (2023) · Puny et al., ICLR (2022) · Cohen & Welling, ICML (2016) · Zaheer et al., NeurIPS (2017) · Satorras et al., ICML (2021)

Equivariance is a constraint on the architecture



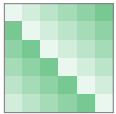
Transform then apply = apply then transform.

- Impose it **layer by layer**: equivariant layers compose into an equivariant network
- Finish with a sum or mean ⇒ invariant output
- The constraint acts on the **weights**: only a subspace of weight matrices is allowed

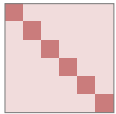
linear layer
$L(x) = W \cdot x$
(n inputs)



no symmetry
n^2 free weights



translations
n weights: a **convolution**



permutations
2 weights: $W = a\, I + b\, \mathbf{1}\mathbf{1}^\top$

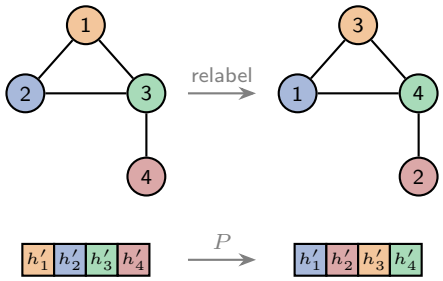
Cohen & Welling, ICML (2016) · Zaheer et al., NeurIPS (2017) · Kondor & Trivedi, ICML (2018)

Which symmetry, which architecture?

Symmetry	Typical data	Architecture	What the constraint does
translations	images, grids, time series	**CNN**	same filter everywhere
permutations (sets)	particles, point clouds	**Deep Sets, transformer**	same function per element, order-blind pooling
permutations (graphs)	molecules, meshes	**GNN**	same message on every edge
E(3): rotations + translations	atoms, 3D point clouds	**EGNN, NequIP, MACE**	vector/tensor features, tensor products
rotations on the sphere	climate, cosmology maps	**spherical CNN**	convolution on the sphere
Lorentz group	particle jets	**Lorentz-equivariant**	four-vectors, Minkowski products
gauge symmetry	lattice field theory	**gauge-equivariant**	built from gauge-invariant loops

Cohen et al., ICLR (2018) · Satorras et al., ICML (2021) · Bogatskiy et al., ICML (2020) · Kanwar et al., PRL 125, 121601 (2020)

Example: a graph neural network is permutation-equivariant



same atoms, new labels:
the outputs are reordered the same way

$$h'_i = \phi\Big(h_i,\ \textstyle\sum_{j\in\mathcal{N}(i)} \psi(h_i, h_j)\Big)$$

- Same ϕ, ψ on every node and edge; a **sum** does not depend on the order of the neighbours
- Relabel the nodes: $H \to PH$, adjacency $A \to PAP^\top$

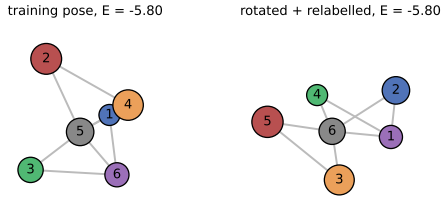
$$F(PH,\ PAP^\top) = P\,F(H, A)$$

- The layer is **equivariant**: each atom keeps its own output (charge, force)
- **Invariance** comes only at the end: $E = \sum_i \rho(h_i)$

Gilmer et al., ICML (2017) · Zaheer et al., NeurIPS (2017) · Bronstein et al., arXiv:2104.13478

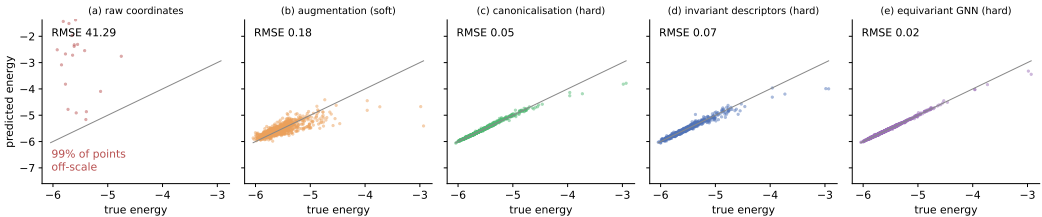
Demo II: one problem, five models

- Energy of a 6-atom cluster (Lennard-Jones + three-body term)
- Trained in **one orientation**, tested on **rotated**, then relabelled, copies
- Same cluster, same energy, 18 different input numbers



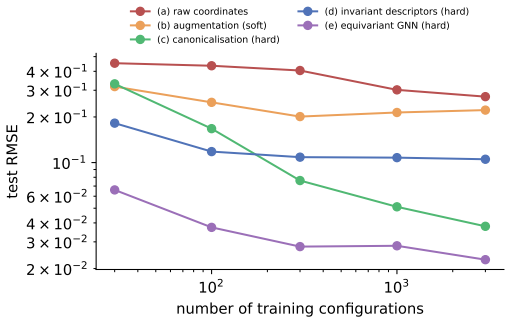
	Input	Network
(a) raw coordinates	18 coordinates	MLP, 2×64 hidden units
(b) augmentation (soft)	18 coordinates, randomly rotated at each step	same MLP, $10\times$ more training steps
(c) canonicalisation	18 coordinates in a standard pose (inertia axes, sorted atoms)	same MLP
(d) invariant descriptors	28 distance and angle functions per atom	same MLP per atom, summed
(e) equivariant GNN	positions; messages along interatomic vectors	one message-passing layer, summed over atoms

Demo II: five ways to handle symmetry

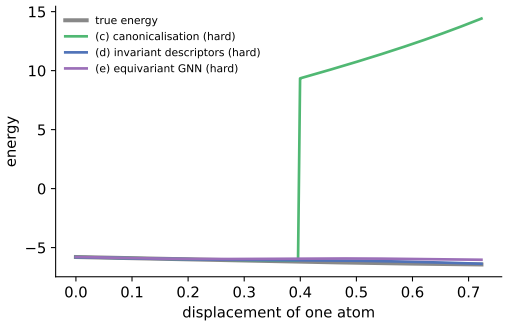


- (a) raw coordinates: off-scale on rotated data
- (b) augmentation: learns the rotations, but with $10\times$ more training; relabel the atoms and it fails
- (c), (d), (e): $E(x) = E(gx)$ to 10^{-14}, by construction; the equivariant network is the most accurate

Demo II: data efficiency and the price of each route



Training data now in random orientations.



Move one atom smoothly: two atoms swap places in the sorting order and the canonicalised model **jumps**.

Hard constraints beyond symmetry

Route	Example
derive from a potential	$\mathbf{F} = -\nabla E_\theta$: conservative forces; $\mathbf{B} = \nabla \times \mathbf{A}_\theta$: divergence-free; Hamiltonian / Lagrangian NNs conserve energy
output parametrisation	softplus $\Rightarrow$ positive; softmax $\Rightarrow$ normalised; $LL^\top \Rightarrow$ positive-definite matrix; input-convex NNs $\Rightarrow$ convex
projection layer	$q_i \leftarrow q_i - (\sum_j q_j - Q)/N$: exact total charge; optimisation layers (OptNet)
trial-function ansatz	$u = g(x) + d(x)\,\mathrm{NN}(x)$: boundary conditions exact (Lagaris et al. 1998)
locality, extensivity	$E = \sum_i \varepsilon(\text{environment}_i)$: size-extensive, linear scaling
hybrid model	keep the known solver, learn only the missing term (universal differential equations)
normalised by construction	normalizing flows: exact probability densities ($\rightarrow$ Part III)

Greydanus et al., NeurIPS (2019) · Cranmer et al., arXiv:2003.04630 · Amos et al., ICML (2017) · Lagaris et al., IEEE TNN 9, 987 (1998) · Rackauckas et al., arXiv:2001.04385

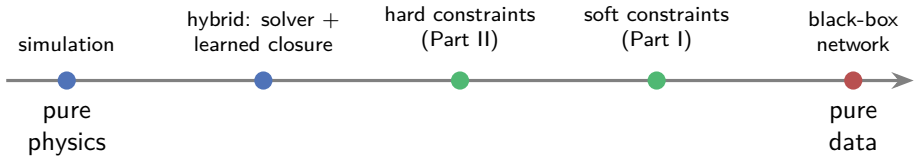
Hard constraints: limitations

- You must know the constraint **exactly**
- **Broken** symmetries: external fields, boundaries, finite-size effects
- Canonicalisation: discontinuities at degenerate configurations
- Invariant descriptors: information loss
- Expressivity and computational cost of equivariant layers
- **Over-constraining**: no room left to learn what the physics misses

Soft vs hard

	Soft (Part I)	**Hard** (Part II)
how	penalty in the loss, augmentation	built into the model
guarantee	approximate, near the data	exact, **everywhere**
data needed	more	less
flexibility	any architecture	needs a suitable design
if the physics is wrong	data can partly correct it	the error cannot be corrected

Where we stand: a spectrum



- The further to the right, the more data you need and the less you can extrapolate
- Most successful AI-for-science models sit in the **middle**
- So far, every model returns **one answer**. What are the other plausible ones, and how much should we trust it? → Part III

III. Generative and probabilistic models

Why generative models in science?

- Parts I and II: learn a **function**, one input $\rightarrow$ one answer
- Now: learn a **distribution** $p(x)$ from examples, the set of *plausible* objects
- Scientific spaces are huge and sparse
  - $\sim 10^{60}$ drug-like molecules; crystals; protein sequences; possible signals
- A generative model learns where the interesting part is, and lets us **explore** it

Generate: propose new candidates
materials, proteins, simulated events

Condition: all answers compatible with what is known
a target property, a measurement $\Rightarrow$ **inverse problems**, **uncertainty**

Score: how plausible is this one?
low probability $\Rightarrow$ **anomaly**, out of distribution

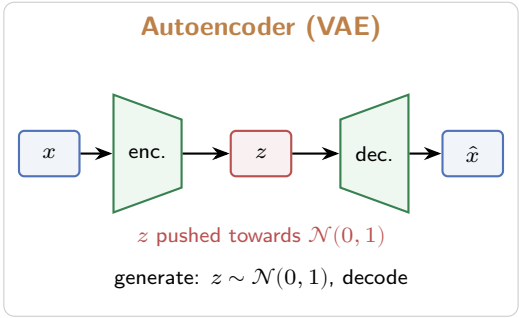
MatterGen: Nature 639, 624 (2025) · RFdiffusion: Nature 620, 1089 (2023) · CaloGAN: PRL 120, 042003 (2018)

A family of generative models

Common idea: draw from a **known distribution**, learn a **map** to the data distribution.

$$z \sim \mathcal{N}(0,1) \xrightarrow{\text{learned map } G_\theta} x = G_\theta(z), \qquad x \sim p_\theta(x) \approx p_{\text{data}}(x)$$

Model	Idea	$p(x)$?
VAE	map from a few Gaussian latent variables	lower bound



Typical use: compact maps, anomaly detection

- All can be **conditioned**: $p(x \mid \text{what we know})$. The demo uses the autoencoder

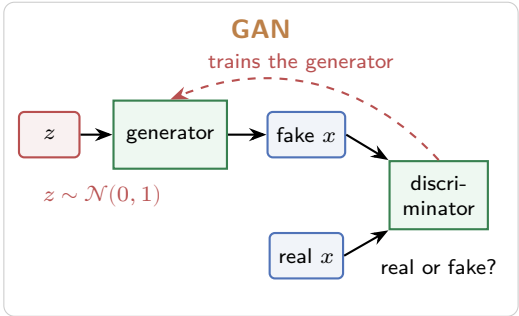
Kingma & Welling, ICLR (2014) · Goodfellow et al., NeurIPS (2014) · Rezende & Mohamed, ICML (2015) · Ho et al., NeurIPS (2020)

A family of generative models

Common idea: draw from a **known distribution**, learn a **map** to the data distribution.

$$z \sim \mathcal{N}(0,1) \xrightarrow{\text{learned map } G_\theta} x = G_\theta(z), \qquad x \sim p_\theta(x) \approx p_{\text{data}}(x)$$

Model	Idea	$p(x)$?
VAE	map from a few Gaussian latent variables	lower bound
GAN	map trained to fool a discriminator	no



Typical use: fast simulation (calorimeter showers)

- All can be **conditioned**: $p(x \mid \text{what we know})$. The demo uses the autoencoder

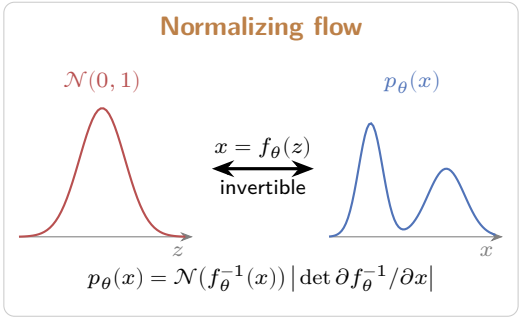
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Flow	the map is invertible	exact



Typical use: exact densities, sampling, inference

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GAN	map trained to fool a discriminator	no
Flow	the map is invertible	exact
Diffusion	the map is many small denoising steps	lower bound

Diffusion model

add a little noise at each step (fixed)



remove it, step by step (**learned**)

generate: start from noise, denoise T times

Typical use: images, molecules, weather ensembles

- All can be **conditioned**: $p(x \mid \text{what we know})$. The demo uses the autoencoder

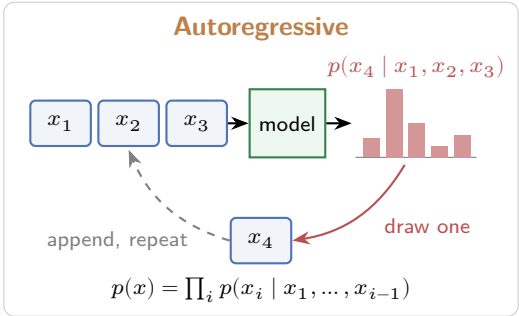
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VAE	map from a few Gaussian latent variables	lower bound
GAN	map trained to fool a discriminator	no
Flow	the map is invertible	exact
Diffusion	the map is many small denoising steps	lower bound
Autoregressive	one element at a time, drawn from a learned discrete distribution	exact



Typical use: text, sequences, LLMs

- All can be **conditioned**: $p(x \mid \text{what we know})$. The demo uses the autoencoder

Kingma & Welling, ICLR (2014) · Goodfellow et al., NeurIPS (2014) · Rezende & Mohamed, ICML (2015) · Ho et al., NeurIPS (2020)

Putting what we know into a generative model

The methods of Parts I and II also apply to generative models.

Soft (Part I)

- Penalise generated samples that violate an equation or a conservation law
- Use a known energy or a simulator as a training signal
- **Guidance**: steer the sampling towards a constraint or a target property

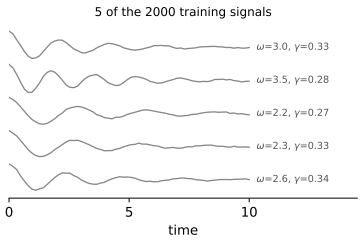
Hard (Part II)

- Equivariant generator $\Rightarrow p(x)$ has the symmetry: a rotated molecule is equally likely
- Valid by construction: positivity, periodicity, fixed composition
- Generate in a physical parametrisation, then map to the data

Examples: equivariant diffusion for molecules, energy-trained flows, physics-guided weather ensembles (GenCast).

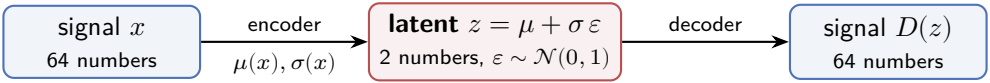
Hoogeboom et al., ICML (2022) · Noé et al., Science 365, eaaw1147 (2019) · Price et al., Nature 637, 84 (2025)

Demo III: a variational autoencoder on signals



2000 damped oscillations $x(t) = e^{-\gamma t}\cos(\omega t)$, 64 samples each, plus noise (0.02). ω and γ are drawn **uniformly**: $\omega \in [2, 4]$, $\gamma \in [0.1, 0.4]$.

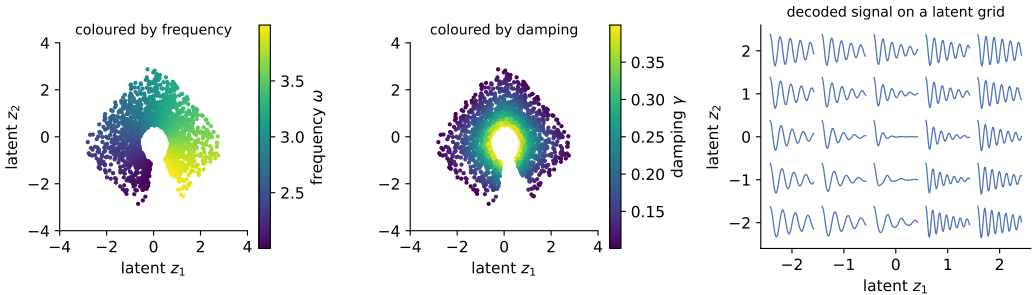
- **Encoder**: MLP $64 \to 256 \to 256 \to 4$. It returns a mean $\mu(x)$ and a width $\sigma(x)$ for each of the 2 latent variables
- **Decoder** D: MLP $2 \to 256 \to 256 \to 64$
- A **standard VAE**, trained on the signals alone: 12000 Adam steps, 20 s on a laptop



$$\mathcal{L} = \underbrace{\frac{\|x - D(z)\|^2}{2\sigma_d^2}}_{\text{reconstruct the data } (\sigma_d = 0.1)} + \underbrace{\mathrm{KL}(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0,1))}_{\text{latent points of the data} \sim \mathcal{N}(0,1)}$$

To generate: draw $z \sim \mathcal{N}(0,1)$, decode. The decoder is the map from $\mathcal{N}(0,1)$ to the data.

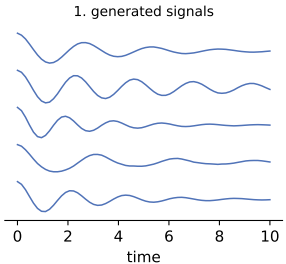
Demo III: the latent space is a map of the signals



The two latent numbers form a (distorted) **map of frequency and damping**, although the model was never given these parameters. Moving in the latent space changes the signal smoothly: this is the space we explore.

Demo III: generate, then check the physics

The model was given no physics. So we test what it generates against what we know.



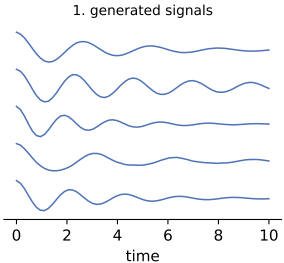
Generate

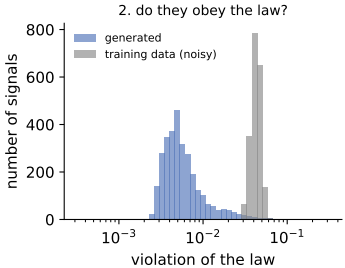
Draw $z \sim \mathcal{N}(0, 1)$, decode.

The signals look right. Are they?

Demo III: generate, then check the physics

The model was given no physics. So we test what it generates against what we know.





Generate

Draw $z \sim \mathcal{N}(0, 1)$, decode.

The signals look right. Are they?

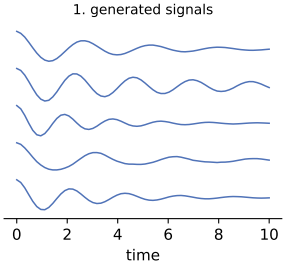
Check 1: the law

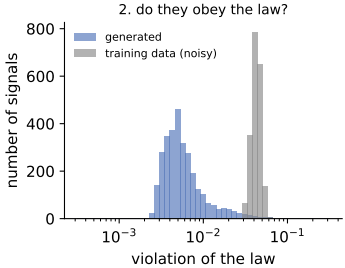
How far is each signal from a solution of $\ddot{x} + 2\gamma\,\dot{x} + \omega_0^2\,x = 0$, whatever ω and γ?

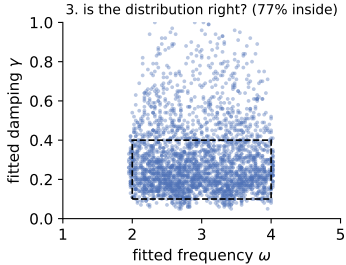
Yes: 8 times closer than the noisy data (0.005 against 0.04).

Demo III: generate, then check the physics

The model was given no physics. So we test what it generates against what we know.







Generate

Draw $z \sim \mathcal{N}(0, 1)$, decode.

The signals look right. Are they?

Check 1: the law

How far is each signal from a solution of $\ddot{x} + 2\gamma\,\dot{x} + \omega_0^2\,x = 0$, whatever ω and γ?

Yes: 8 times closer than the noisy data (0.005 against 0.04).

Check 2: the distribution

Fit ω, γ to each signal. Are they in the training ranges (dashed)?

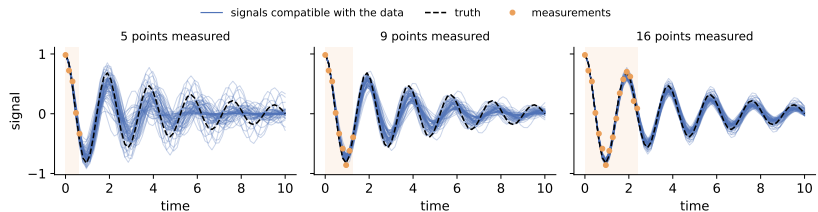
Partly: 77% are. The others come from the hole in the latent map.

The **form** of the signals is learned well, their **distribution** only approximately.

Uncertainty: the generative model as a prior

A few noisy measurements y of a signal. What is the rest? Every signal that is

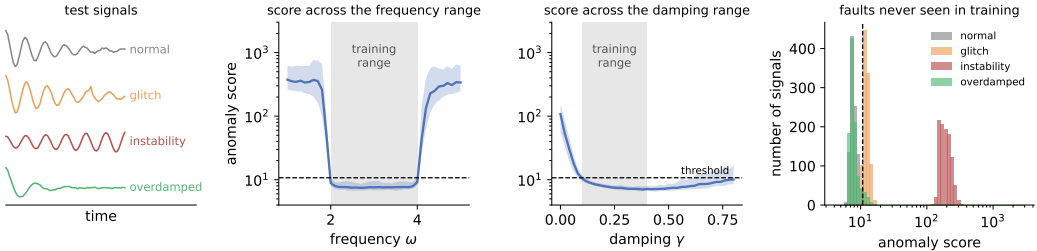
$$p(x \mid y) \propto \underbrace{p(y \mid x)}_{\text{compatible with the data}} \times \underbrace{p(x)}_{\text{physical: the generative model}}$$



- The **prior** $p(x)$ can be **pretrained** once (simulations, archives) and reused
- Spread of the samples = **uncertainty**
- **Limits**: depends on the prior; unreliable for cases unlike the training data

Anomaly detection: low-probability events

Train on **normal** data only; score a new signal by $-\ln p(x)$. No example of a fault is needed.



- The score finds the **edge of the training data** in frequency; 93–100% of the faults are flagged
- Out of distribution does not mean faulty: $\omega = 4.5$ is a valid oscillation
- Pitfall: overdamped signals, outside the training range, pass as normal (7% flagged)

Nalisnick et al., ICLR (2019) · Chandola, Banerjee & Kumar, ACM Comput. Surv. 41, 15 (2009)

Probabilistic models: links and pitfalls

Links to this week

- **Anomaly detection**: low likelihood under a model of "normal" $\Rightarrow$ suspicious
- **Optimisation**: uncertainty decides where to evaluate next (Bayesian optimisation)
- **Surrogates**: error bars tell you when to stop trusting the model (out of distribution)

Limitations

- Generative models are not easy to evaluate: quality of samples is not enough
- Mode collapse; rare events under-represented
- Overconfidence out of distribution
- Uncertainty that is itself not calibrated

Other ways to get an uncertainty

- **Ensembles**: train several models, use their spread; **Gaussian processes**; Bayesian neural networks, MC dropout
- Predict a width $\sigma_\theta(x)$ together with the value
- **Conformal prediction**: intervals calibrated on held-out data, for any model

Conclusion

Three ways to put physics into a model

	encodes	guarantee	tools	use when
I. Soft	equations, limits, sum rules in the loss	approximate, near the evaluated points	PINNs, penalties, augmentation	physics known but approximate; few data
II. Hard	symmetries and structure in the model	exact, everywhere	invariant features, canonicalisation, equivariant nets, potentials	constraint known exactly
III. Probabilistic	the full distribution	error bars (if calibrated)	GPs, ensembles, flows, diffusion	you need uncertainty or samples

Resources

Software

- General: PyTorch, JAX, scikit-learn
- Physics-informed: DeepXDE, neuraloperator
- Symmetry: PyTorch Geometric, e3nn, MACE
- Uncertainty: GPyTorch, BoTorch
- Generative: sbi (simulation-based inference)

Demos of this lecture: `demos/` (three notebooks, CPU, about 1 min each)

To go further

- *Physics-based Deep Learning* (Thuerey et al., online book)
- *Geometric Deep Learning* (Bronstein et al., 2021)
- *Deep Learning* (Goodfellow, Bengio, Courville)

Next: Accelerators and AI

Questions?

Have a great hackathon!